

- Q.25 Explain the function of Squelch circuit used in communication receiver.
- Q.26 What do you mean by radio receiver? List the functions of radio receiver.
- Q.27 Draw the block diagram of AM receiver.
- Q.28 Write a note on half wave dipole.
- Q.29 Explain the radiation principle of antenna.
- Q.30 Define the terms associated with antenna - radiation pattern and gain.
- Q.31 Explain electromagnetic spectrum and its various ranges.
- Q.32 Explain ground wave propagation.
- Q.33 Discuss the term skip distance.
- Q.34 What is active satellite? Explain its functions.
- Q.35 Write advantages and disadvantages of satellite communication.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write short note on the following:
a) Fidelity b) Sensitivity
- Q.37 Draw and explain the block diagram of FM super heterodyne radio receiver.
- Q.38 What are the different modes of radio wave propagation? Explain sky wave propagation.

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SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Pre-Emphasis circuit is used to amplify
a) Low frequency b) High frequency
c) Both a and b d) None of these
- Q.2 Long wave AM broadcast transmitters need
a) Small carrier power
b) Medium carrier power
c) Very large carrier power
d) No need for carrier power
- Q.3 The standard IF value for FM receivers is
a) 455 kHz b) 455 MHz
c) 10.7kHz d) 10.7MHz
- Q.4 RF circuit in super heterodyne radio receiver is used to
a) select the wanted signal
b) reject the interference
c) Reduce the noise figure
d) All of above

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- Q.5 The current distribution on a half wave dipole is
 a) Uniform b) Sinusoidal
 c) Triangular d) Complex
- Q.6 The reflector in a Yagi antenna
 a) Is longer than the driven element
 b) Is shorter than the driven element
 c) Can be longer or shorter than the driven element
 d) Does not exist
- Q.7 The ionosphere extends from earth surface to a height of
 a) 18km b) 50km
 c) 100km d) 400km
- Q.8 The skip distance for radio wave increases with
 a) Increase in frequency
 b) reduction in frequency
 c) temperature of atmosphere
 d) None of above
- Q.9 Frequency range of LF band.
 a) 30-300 MHz b) 30-300 kHz
 c) 300-3000 kHz d) 3-30 Mhz
- Q.10 Various components of VSAT are
 a) A dish antenna b) A transceiver
 c) A satellite route d) All of above

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SECTION-B

- Note:** Objective type questions. All questions are compulsory. (10x1=10)
- Q.11 Armstrong modulation is _____ method of FM generation.
- Q.12 A microphone converts electrical signal into sound. (T/F)
- Q.13 FET stands for field effect transistor. (T/F)
- Q.14 The process by means of which information is extracted from the modulated signal is called _____.
- Q.15 The radiation pattern of dish antenna is unidirectional /bidirectional.
- Q.16 Antenna beam width is a measure of directivity of an antenna. (T/F)
- Q.17 What is the full form of VHF?
- Q.18 VSAT stands for _____.
- Q.19 Full form of AGC _____.
- Q.20 Define the term perigee.

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 Classify the AM transmitter in the basis of type of modulation involved.
- Q.22 Draw the block diagram of reactance FM transmitter.
- Q.23 Define the term selectivity.
- Q.24 What is AGC? Differentiate between simple and delayed AGC.

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